



D3.1 – Battery Electric Vehicle Traction Motor Composition Data Description

WP3 – Task 3.1
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Artificial Intelligence (AI) was used to enhance the clarity of certain aspects of this document. The AI tool used in this document was ChatGPT. It was used in sections 1 to 4. The purpose was to rephrase and enhance the clarity of certain sentences. The authors retain full accountability for the content and its accuracy.

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Harmonized datasets for secondary RM sources for the twin transition – Version 1 (29.05.2026)

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Summary

The dataset presented in this document is part of Deliverable 3.1 and provides information on the consolidated material composition dataset for BEV traction motors.

Keywords

Battery electric vehicles, material compositions

Abbreviations and acronyms

Acronym	Description
AI	Artificial Intelligence
BEV	Battery Electric Vehicle
CO ₂	Carbon Dioxide
EEA	European Environment Agency
EESM	Electrically Excited Synchronous Machine
EV	Electric Vehicle
HEV	Hybrid Electric Vehicle
IM	Induction Motor
PM	Permanent Magnet
PMSM	Permanent Magnet Synchronous Motor
PHEV	Plug-in Hybrid Electric Vehicle
RM	Raw Material
SUV	Sport Utility Vehicle
WP	Work Package

1 Introduction

This dataset provides consolidated composition data for battery electric vehicle (BEV) traction motors considering three technologies – permanent-magnet (PM) containing traction motors, induction motors (IM) and electronically excited synchronous machines (EESM). BEVs are classified into 12 segments.

2 Grouping of products (product keys)

The aim of this data collection is to provide traction motor component and material weights by car segment and motor topology, in alignment with a stock-and-flow model that differentiates BEVs at the segment level. To ensure consistency with this modelling approach, component weights are reported at the vehicle level rather than per individual motor. Table 2 shows product codes for car models considered derived from Table 1.

Table 1: Product keys

Code	Category	Example
V	Product	Vehicle
03	Powertrain type	elvBEV (03), elvPHEV, elvHEV, elvPetrol, elvDiesel
01	Vehicle type	Car (01), Bus, Train
03	Fuel type	electric (03), diesel, petrol
01--02	Body type	SUV (J) or Hatchback/Sedan
01--06	Car segment	A, B, C, D, E, F

Table 2: Product groupings

Seg.	Code	Typical Name	Example Car Models
A	V0301030101	Mini / City Car	Fiat 500, Hyundai i10, Kia Picanto
B	V0301030102	Small / Subcompact	Volkswagen Polo, Ford Fiesta, Renault Clio
C	V0301030103	Compact / Small Family Car	VW Golf, Toyota Corolla, Ford Focus
D	V0301030104	Mid-size / Large Family Car	BMW 3 Series, Mercedes-Benz C-Class, Audi A4
E	V0301030105	Executive / Large Sedan	BMW 5 Series, Mercedes-Benz E-Class, Audi A6
F	V0301030106	Luxury / Flagship Car	Mercedes-Benz S-Class, BMW 7 Series

3 Product/component structure

A clear product model was defined by setting system boundaries that distinguish the core motor (stator, rotor, windings, magnets) from ancillary components such as housing and gearbox. The product was then decomposed into a standard component hierarchy based on industry and teardown knowledge, ensuring consistency with typical motor designs. This structure was adapted to the project's material focus by disaggregating components containing critical raw materials (e.g., rare earth elements in magnets, copper in windings) while aggregating less relevant parts. The housing and cooling system was modelled as a single component, as both are predominantly composed of aluminium and are often structurally integrated.

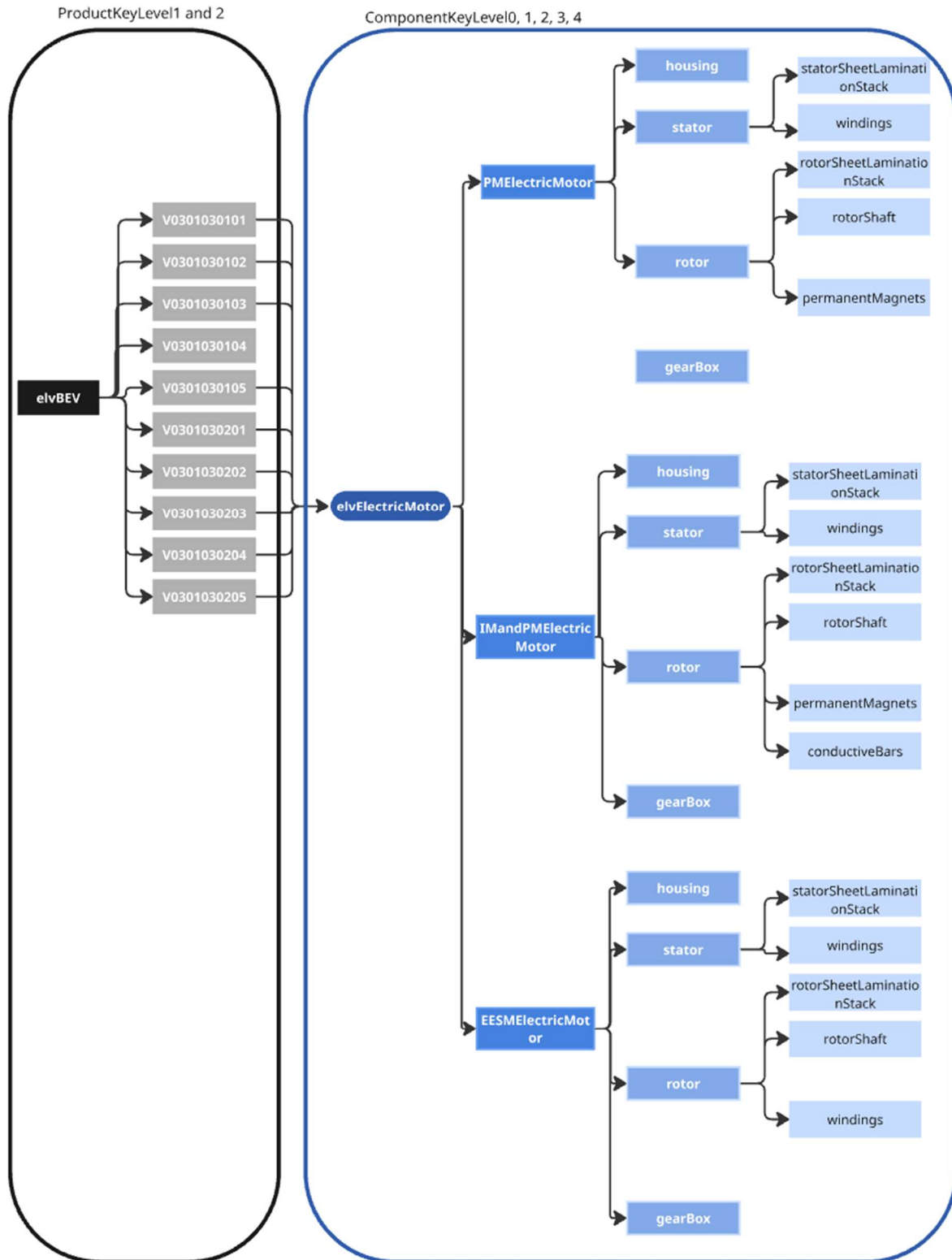


Figure 1: Product component structure of the dataset

4 Market share information and market structure

The market for BEVs is a fragmented market (low four-firm concentration ratio). The market is split among a large number of competitors with relatively small individual shares. Within each

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segment individual car models are assumed to have uniform weight. Market shares are derived from data on new car registrations collected from the European Environmental Agency (EEA, 2025).

5 Data sources and consolidation approach

Component weight-torque relationships were derived by extracting specific data points from literature and fitting a regression line. Uncertainty is defined by the regression confidence interval. Specific data is derived from Drexler et al. (2025) to predict component mass by torque using linear regression. Data on car models and their respective torques are collected from the EV Database and classified into segments. Monte Carlo simulation is applied to propagate this uncertainty.

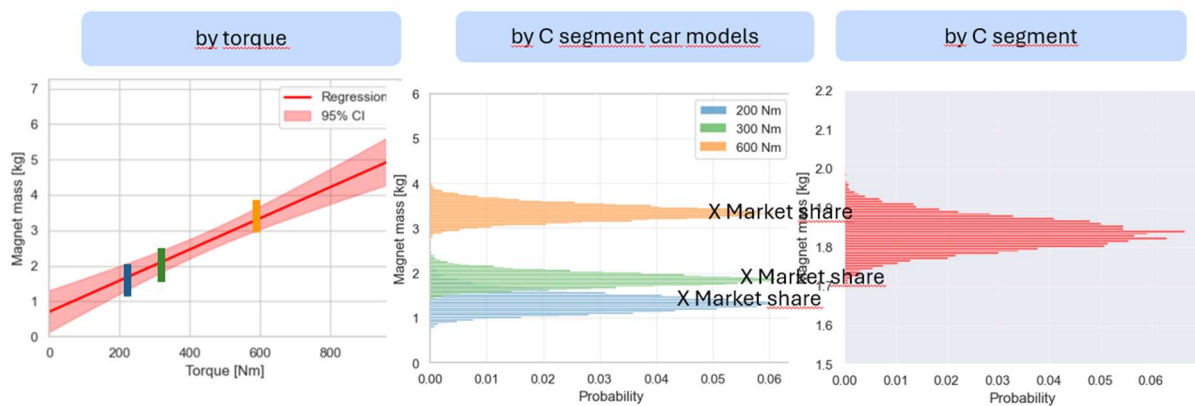


Figure 2: Data consolidation steps

For each segment, component weights are sampled from the regression uncertainty (confidence interval) for each torque value. Sampled weights are assigned to each vehicle model based on its torque. All models within a segment are weighted equally. For each iteration, weights are averaged across all models within a segment. The consolidation steps are illustrated in Figure 2.

6 Contributors

Rebeka Anspach – Data structuring and consolidation

Matthias Rösslein – Conceptualization of the data consolidation, review

Biborka Boga – Conceptualization of the data consolidation, review

Lukas Feldmann – Review

Harald Desing – Review

Kirsten Remmen – Coordination

Sophie Personnaz, VALEO – Review of the data

7 References

Data sourced from the following references were used to derive consolidated data points:

Drexler, D., Kampker, A., Born, H., et al. Advances in electric motors: a review and benchmarking of product design and manufacturing technologies. *Elektrotech. Inftech.* 142, 312–345. <https://doi.org/10.1007/s00502-025-01331-3> (2025).

EEA. *Monitoring of CO₂ emissions from passenger cars, 2024 – Provisional data* [Data set]. <https://doi.org/10.2909/5814203d-f36d-4fee-90d4-64bb348d15a6> (2025).

EV Database. EV Database: All electric vehicles in Europe <https://ev-database.org/>. Online database of electric vehicle specifications. (2026).